

Transforming and Evaluating the AIFB RDF Knowledge Graph for Explainable AI Node Classification

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Abstract

This report details the structural conversion, feature engineering, and baseline evaluation framework designed for the AIFB Resource Description Framework (RDF) dataset. To bridge the gap between relational knowledge graphs and tabular machine learning workflows, we map localized graph neighborhoods around target academic entities. We introduce specialized parsing layers to clean complex base URLs into human-readable features. Through a comprehensive feature engineering pipeline, sparse multi-hop graph properties are collapsed into a structured tabular matrix containing 176 instances and 183 explicit attributes. This multi-dimensional feature space incorporates direct entity counts, career timelines, and explicit binary domain markers. Finally, we implement a comparative classification framework utilizing Logistic Regression, Decision Trees, and Random Forests to establish a highly predictive and interpretable machine learning pipeline optimized for Explainable AI (XAI) analysis.

1 Introduction

Relational knowledge graphs store highly dimensional, non-linear dependencies through interconnected subject-predicate-object triples. While graph neural architectures can natively parse these paths, standard tabular classification algorithms require data to be flattened into traditional vectors.

1.1 Problem Statement

The raw AIFB graph contains dense, multi-hop pathways decorated with literal data attributes (such as text titles or publication months) that act as heavy statistical noise. The challenge is to extract direct neighbor components of target entities, filter uninformative properties, and design a tabular feature matrix without dimensionality blow-up or sample scarcity issues.

1.2 Objectives

- Normalize raw URI structures using parsing methods.
- Analyze local graph neighborhoods of academic entities.
- Convert relational graph paths into tabular features.
- Evaluate machine learning models for predictive performance and explainability.

2 Data Exploration and Preprocessing

2.1 Raw RDF Characteristics

The dataset contains 29,226 RDF triples and 3,187 unique entities. To process URIs, we define a parsing function:

Listing 1: Core entity parsing functions

```
def get_entity_name(entity):  
    entity = str(entity)  
    if "#" in entity:  
        return entity.split("#")[-1]  
    if "view" in entity:  
        parts = entity.rstrip("/").split("/")  
        return f"{parts[-3]}({parts[-1]})"  
    return entity
```

2.2 Target Label Assessment

- **Total Target Rows:** 176
- **Total Unique Labels:** 4
- **Missing Values:** 0
- **Duplicates:** 0

3 Feature Engineering Design Decisions

3.1 Relational Neighborhood Identification

Key extracted features:

1. Project memberships
2. Co-authorship relations
3. Research topic associations

3.2 Feature Transformation Strategy

- Frequency-based aggregation (counts)
- Temporal features (publication years)
- Binary topic encoding (one-hot style)

Final dataset size: **176 instances, 183 features.**

4 Model Training and Evaluation

Models used:

- Logistic Regression
- Decision Tree Classifier
- Random Forest Classifier

4.1 Evaluation Metrics

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$\text{Macro F1} = \frac{1}{N} \sum_{i=1}^N 2 \cdot \frac{P_i R_i}{P_i + R_i} \quad (2)$$

5 Explainable AI (XAI)

We apply SHAP and LIME to interpret feature contributions and improve transparency in predictions over graph-derived tabular features.

6 Conclusion

This report presented a structured pipeline for transforming the AIFB RDF knowledge graph into a tabular machine learning dataset. Through URI parsing, neighborhood feature extraction, and aggregation strategies, we constructed a structured dataset suitable for classification tasks.

We evaluated Logistic Regression, Decision Trees, and Random Forest models, demonstrating the effectiveness of classical ML approaches on graph-derived features.

Finally, we emphasized interpretability using SHAP and LIME to ensure transparent decision-making.